Does Idiomatic Phrasing Degrade LLM Task Performance? A Controlled Study Across Instruction-Tuned Decoder Models

Lihu Zur (322216151), Assaf Haik Barouch (206783441)

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1 Introduction

Idiomatic expressions — those whose meaning cannot be derived compositionally from their individual words, e.g., *"spill the beans"* or *"par for the course"* — are routinely treated in NLP as a harder case for language models than direct, literal phrasing. The premise is intuitive: idioms are culturally specific and comparatively rare in balanced training data relative to literal phrasing of the same content. It is, however, rarely tested *in isolation*: any two rewordings of a sentence already differ lexically and syntactically, so an observed accuracy drop after "idiomatizing" it could simply be an artifact of rewording in general.

Central research question: does phrasing a task input idiomatically cause a measurable degradation in a language model's task accuracy, *over and above* the degradation that any comparable paraphrase (a non-idiomatic rewording) already causes?

To answer this cleanly, we rewrite every original example x into two controlled variants that share the same rewriting budget but differ in exactly one property: $x_{\text{paraphrase}}$, reworded **without** idioms, the *control* condition isolating the effect of rewording alone; and $x_{\text{idiomatic}}$, reworded **with** one or more natural idiomatic expressions and the same meaning. The label y is preserved across all three variants by construction (§2.2), and each variant is evaluated under an identical prompt template, so only the input's surface form changes between runs. The question matters practically: idiomatic language is common in the user-generated text deployed systems must handle. If idiomaticity has an isolable negative effect, that bears on prompt engineering, augmentation strategy, and robustness evaluation; if apparent idiom failures are merely generic paraphrase sensitivity, effort spent specifically on idiom handling is misplaced relative to general robustness-to-rewording work.

Contributions. We contribute (i) the paraphrase-controlled design just described, in which the paraphrase arm absorbs the generic cost of rewording and leaves the idiom-specific effect isolated; (ii) a *sole-cause* diagnostic that, restricted to items a model already answers correctly, identifies which single rewrite is responsible when exactly one of the two flips the answer; (iii) dual significance testing, pairing a continuity-corrected McNemar test with an independent non-parametric bootstrap to guard against the chi-square approximation on small discordant-pair counts; (iv) breadth — 17 instruction-tuned decoder LLMs from 0.5B to 9B across 11 families, on three datasets spanning sentiment, knowledge QA and inference, for 51 runs with a pooled cross-model test per dataset; and (v) a reproducible four-stage pipeline with LLM-judge validation of every augmented example and per-artifact provenance (§2).

1.1 Related Work

Idiom processing in NLP. Prior work has mostly focused on *detecting* or *translating* idioms rather than measuring their effect on downstream accuracy. Tayyar Madabushi et al. [1] probe pretrained models on the literal-versus-idiomatic distinction, finding they do reasonably well given one or a few labelled examples per expression but degrade substantially zero-shot on unseen expressions, and Chakrabarty et al. [2] introduce FLUTE, a 9,000-instance figurative-language benchmark spanning idioms, similes, metaphors, and sarcasm. The LLM era has largely inherited this framing: Phelps et al. [3] find prompted LLMs competitive but still short of fine-tuned task-specific models on three idiomaticity datasets even at GPT-4 scale, and Mi et al. [4] show with a controlled contrastive dataset that LLMs frequently fail to resolve idiomaticity when doing so depends on context, arguing that headline benchmark success may rest on reasoning shortcuts. All of this measures idiom *comprehension* — establishing it as a genuine weak point — rather than what idiomatic phrasing costs on a task that is not itself about idioms.

Figurative language inside a task. Closer to our question is work that scores figurative material within a downstream task. Stowe et al. [5] pair idiomatic and metaphoric sentences with literal counterparts and find that MNLI-finetuned RoBERTa reliably recognizes figurative–literal entailment yet fails on matched non-entailing pairs. For metaphor, Sanchez-Bayona and Agerri [6] conclude from NLI and QA experiments that LLM performance is shaped more by lexical overlap and sentence length than by metaphorical content — an independent finding consonant with ours. Our design departs from these in introducing the figurative property by a controlled rewrite of items whose original form is also scored, rather than annotating it where it already occurs, which is what makes a matched control arm possible.

Why a paraphrase control is necessary. Models are sensitive to input edits that leave the correct answer intact: Ribeiro et al. [7] flip predictions by *rewording* inputs under semantics-preserving rules (SEARs), and for LLMs the effect is larger still — Cao et al. [8] report a 45.5-point gap between the best and worst semantically equivalent phrasings of the same input. A bare original-versus-idiomatic contrast is therefore uninterpretable on its own, because it measures idiomaticity confounded with generic rewording sensitivity.

Controlled evaluation design. Both our rewrites are invariance tests in the sense of CheckList [9] — label-preserving perturbations under which the prediction ought not to change. They are deliberately the mirror image of contrast sets [10], which perturb minimally in order to *change* the gold label. What we add is subtraction: reporting the gap between a treatment and a matched control condition instead of the treatment alone, the move that control tasks make for probing classifiers [11]. Control conditions have appeared in figurative evaluation before — MUNCH includes deliberately inapt paraphrases to detect lexical-similarity shortcutting [12] — but as a distractor in the answer space, not as a matched arm whose accuracy is subtracted. Concurrent work argues the general case directly: Yang et al. [13] observe that any counterfactual edit bundles the factor of interest with incidental surface variation, and report that on MedQA a 14.9% flip rate under a demographic edit is statistically indistinguishable from the 14.1% induced by paraphrasing alone. We take that logic as settled and instantiate it for a linguistic intervention; unlike the demographic case, a residual effect does survive here on inference. Studies that inject a property without such a baseline, such as the dialect rewrites of GLUE by Ziems et al. [14], accordingly bound its contribution from above rather than estimating it. To our knowledge idiom-inserted rewrites of general-purpose NLU benchmarks have not previously been scored against a matched, semantics-preserving paraphrase arm over the same items, which is what licenses attributing the residual to idiomaticity rather than to rewording.

2 Methodology

2.1 Pipeline overview

The project is implemented as four independently re-runnable stages (Figure 1) that communicate only through versioned files on disk, plus YAML configs that set each stage's run parameters; dataset-specific loading and answer-parsing logic lives in per-dataset modules. Any stage can therefore be re-executed alone — evaluating a new model does not require repeating cleaning or augmentation — and every artifact records the config hash and model/prompt version that produced it.

Stage 1: Clean. Each raw dataset is loaded from Hugging Face and reduced to one clean CSV of rows suitable for rewriting: rows outside a configured length band are dropped (very short inputs leave no room for an idiom; very long ones are expensive to augment and dilute the measured effect), text is normalized (Unicode NFC, whitespace collapsing), exact duplicates are removed, and every retained row gets a stable identifier that aligns all its downstream variants. The label y and dataset-specific metadata (e.g., MMLU's answer choices) carry through unchanged.

2.2 Augmentation into paraphrase and idiomatic variants

Stage 2: Augment. Each cleaned row is rewritten by an external LLM (Google Gemini) into two variants using two distinct, fixed prompt templates. Templates are selected per dataset, so a task whose input is not a free-standing sentence can use wording suited to it (MNLI, whose rewrite target is a premise judged against a fixed hypothesis, uses its own NLI-specific pair of templates) without perturbing the prompt hashes, and therefore the warm caches, of the other datasets:

the *paraphrase prompt* asks for a rewording in different words that explicitly avoids idiomatic expressions, and the *idiomatic prompt* asks for one that naturally incorporates one or more idioms or figures of speech while preserving meaning, register, and length. Both are given the task label y in the prompt itself, so a rewrite cannot silently invert the correct answer, e.g., flipping a positive sentiment snippet into a negative one. An external, deliberately stronger LLM is used rather than one of the models under evaluation, since the augmenter's own idiom/paraphrase judgment must be more reliable than the systems being tested.

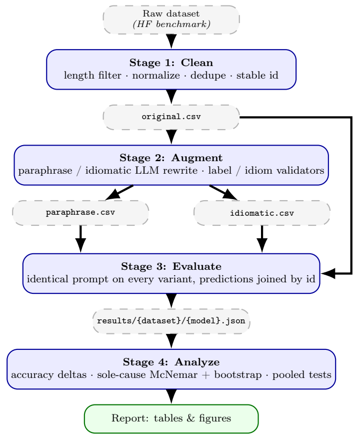

Figure 1: The four-stage pipeline (the stage orchestrators carry no dataset- or model-specific branching; each dataset adds its own loader and evaluator module). The same pipeline instantiates once per dataset/model combination. Raw data is cleaned (Stage 1), rewritten into validated paraphrase and idiomatic variants (Stage 2), evaluated per model across every variant with an identical prompt (Stage 3), and aggregated into the accuracy-delta and sole-cause significance analysis (Stage 4). `original.csv` feeds both Stage 2 (as the rewriting source) and Stage 3 (as the control variant) directly.

Every augmented row then passes a validator chain: **label preservation**, an LLM-judge check that the rewrite has not altered the correct label; and **idiom presence/absence**, a per-variant LLM-judge check requiring a genuine idiom in the idiomatic variant and none in the paraphrase. Rows that fail validation after a bounded number of retries are dropped from that variant (recorded in a sidecar manifest for transparency), which is why the paraphrase/idiomatic CSVs are typically somewhat smaller than the original CSV. An on-disk response cache, keyed by (prompt hash, augmenter model, input id), makes reruns idempotent and avoids re-incurring API cost for rows that have already been successfully augmented under an unchanged configuration. Table 1 shows one produced triple per dataset.

2.3 Datasets

Three datasets are evaluated, each scored with the accuracy metric used in its original publication. Stage 2 rewrites one field per dataset — SST-2's sentence, MMLU's question stem, MNLI's premise — and everything else, including MMLU's answer choices and MNLI's hypothesis, stays byte-identical across the three variants.

SST-2 [15] is binary sentiment classification over short movie-review snippets; **MMLU** [16] is multiple-choice knowledge questions spanning 57 subjects; and **MNLI** [17] is three-way inference (*entailment* / *neutral* / *contradiction*) over premise–hypothesis pairs from ten written and spoken genres.

The evaluated split differs by dataset according to what each release makes available with labels: MMLU uses its `validation` split and MNLI its two validation splits (below), while SST-2, whose GLUE test labels are withheld, draws from `train` and `validation` combined, capped at 2,000 rows.

The three cover qualitatively different task types under one protocol: short, informal, already-somewhat-figurative text; longer, formal, fact-oriented multiple-choice questions; and sentence-pair inference. MNLI is the most direct test of the three, because in sentiment or knowledge QA an idiom perturbs the surface of an input whose answer is otherwise independent of it, whereas an entailment judgement turns on a literal reading of the premise itself. Its three-way label space also sets chance at 33% rather than 50%, which depresses the per-row rate of sole-cause events relative to a binary task.

MNLI-specific protocol. MNLI pairs roughly three hypotheses with every premise; emitting all of them would repeat the same premise and violate the independence assumption behind the paired test, so the loader keeps only the first hypothesis per premise. Rows come from the two validation splits combined (matched, 9,815 pairs; mismatched, 9,832), the test labels being withheld. Because the mismatched half contributes the five genres absent from training, each row's genre is recorded so that genre shift stays visible as a confound. After first-hypothesis selection and cleaning, 5,127 premises survive to form the MNLI original file, of which 4,709 (91.8%) additionally clear validation in *both* rewritten variants and so form the aligned triples the analysis runs on.

Table 1: One aligned triple per dataset, verbatim from `datasets_out/`. Only one field per dataset is rewritten — SST-2's sentence, MMLU's question stem, MNLI's premise — and the label and every other field are byte-identical across the three rows. Each example is a *sole-cause* case: the models counted answered the original and the paraphrase correctly and the idiomatic variant incorrectly, so the idiom alone broke the answer. **Bold** marks the aligned span each variant puts in the same role, so the three rows can be read against one another. Note that in each, the meaning is preserved while the surface requires an extra resolution step — the MNLI idiom pairs a negation with *sell short*, and the SST-2 idiom carries an apparently positive phrase in a negative-sentiment snippet.

MNLI — hypothesis: *"No one would give them credit like that."*; gold label *contradiction*. Idiomatic-only-wrong for 14 of 17 models.	
original	Even the Scots are **likely to give them more credit** than that.
paraphrase	Even the residents of Scotland **will probably accord them a higher level of recognition** than that amount.
idiomatic	Even the Scots are **unlikely to sell them short** like that.
SST-2 — gold label *negative*. Idiomatic-only-wrong for 12 of 17 models.	
original	in which the rest of the cast was **outshined** by ll cool j.
paraphrase	in which the rest of the cast was **outperformed** by ll cool j.
idiomatic	in which LL Cool J **stole the show** from the rest of the cast.
MMLU — gold answer *"The nation invests in research and development of new technology."*; the four choices are untouched. Idiomatic-only-wrong for 9 of 17 models.	
original	Which of the following scenarios would **increase** a nation's production possibility frontier (PPF)?
paraphrase	Which of the following situations would **expand** a country's production possibility frontier (PPF)?
idiomatic	Which of the following scenarios would **push out** a nation's production possibility frontier (PPF)?

2.4 Evaluated models

Rather than a single-model case study, we evaluate a broad panel of 17 open, instruction-tuned decoder LLMs spanning 0.5B to 9B parameters across 11 distinct model families and pretraining lineages: the Qwen2.5-Instruct family (0.5B / 1.5B / 3B / 7B), the Gemma-2-it family (2B / 9B), the Falcon3-Instruct family (3B / 7B), the OLMo-2-Instruct family (1B / 7B), Mistral-7B-Instruct-v0.3, Phi-4-mini-instruct (3.8B), Granite-3.1-2B-Instruct, H2O-Danube3-4B-Chat, Yi-1.5-6B-Chat, SmolLM2-1.7B-Instruct, and StableLM-2-1.6B-Chat. Spanning multiple scales *within* a family (four Qwen2.5 sizes) as well as different families at similar scale lets the analysis separate size-driven from family-driven effects rather than conflating them, as a single family swept across sizes would. All models are evaluated zero-shot with an identical dataset-specific prompt template applied to all three variants of a row, so only the input text differs between runs; a device-agnostic runner selects CUDA, Apple-Silicon MPS, or CPU and adjusts precision accordingly.

Stage 3: Evaluate. For each (model, dataset) pair, the model is run on all three variant CSVs with the identical prompt, and predictions are joined by row id, yielding one matched (original, paraphrase, idiomatic) triple of predictions per surviving row. Per-variant accuracy is computed directly from these joined predictions.

2.5 Statistical framework

Stage 4: Analyze. Two families of statistic are computed. The first is per-variant accuracy and the two deltas between variants, reported throughout in percentage points (pp):

$$\Delta_{\text{par}} = \text{acc}_{\text{par}} - \text{acc}_{\text{orig}}, \qquad \Delta_{\text{idi}} = \text{acc}_{\text{idi}} - \text{acc}_{\text{par}}. \tag{1}$$

Δ_{par} measures the cost of rewording and Δ_{idi} the additional cost of idiomaticity on top of it; the research question of §1 is precisely whether $\Delta_{\text{idi}} < 0$. Each is accompanied by a paired-bootstrap confidence interval and a McNemar exact p-value over the corresponding discordant pairs.

The second, and our primary diagnostic, is a *sole-cause* test, which attributes degradation to a single rewrite type rather than to ordinary noise. It restricts attention to rows where all three variants were produced and the model answered the **original** correctly — so any subsequent error is attributable to the rewrite, not to the item being hard for that model — and then counts b = *idiomatic-only-wrong* (the idiomatic rewrite alone broke an otherwise-correct answer) and c = *paraphrase-only-wrong* (the paraphrase rewrite alone broke it). Rows where both rewrites agree, breaking it or not, carry no information about which is worse and are excluded, exactly as in any paired test. The gap between b and c is then assessed two ways:

McNemar's test [18], the standard test for paired nominal before/after comparisons, with the continuity correction of Edwards [19]:

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c}, \qquad \text{df} = 1, \tag{2}$$

Table 2: Per-dataset summary over the aligned variant triples. Δ values are means across models in percentage points; *sig.* counts models with a significant negative / positive Δ_{idi} ($p < 0.05$, McNemar exact). *Parse-clean* repeats both after dropping every triple in which any variant failed to parse, since an unparseable generation is scored as incorrect. The pooled sole-cause columns sum b (idiomatic-only-wrong) and c (paraphrase-only-wrong) across models and test the gap with continuity-corrected McNemar and a 10,000-resample paired bootstrap.

			All triples			Parse-clean		Pooled sole-cause			
Dataset	Models	n	Δ_{par}	Δ_{idi}	sig.	Δ_{idi}	sig.	b / c	χ^2	p_{χ^2}	p_{boot}
SST-2	17	1,548	−1.25	+1.31	0 / 5	+0.67	0 / 3	665 / 887	31.5	<0.0001	<0.0001
MMLU	17	1,085	−0.07	−0.71	1 / 0	−0.73	1 / 0	578 / 468	11.4	0.0008	0.0012
MNLI	17	4,709	−0.69	−1.16	10 / 0	−1.14	10 / 0	3351 / 2751	58.8	<0.0001	<0.0001

Table 3: Per-model Δ_{idi} (pp) on each dataset, ordered by parameter count and split into two panels. **Bold** marks Δ_{idi} significant at $p < 0.05$ (McNemar exact); † marks models whose sole-cause gap between b and c is significant on *both* the McNemar and bootstrap tests, in whichever direction. Negative values support the hypothesis that idiomatic phrasing costs accuracy beyond paraphrasing; note that the SST-2 columns are almost entirely positive.

Model	SST-2	MMLU	MNLI	Model	SST-2	MMLU	MNLI
Qwen2.5-0.5B	+0.45	−1.75	−0.40	Phi-4-mini-3.8B	−0.19	−1.84	**−1.51**†
OLMo-2-1B	+1.29	−0.28	**−1.70**†	H2O-Danube3-4B	**+1.74**	**−2.49**	+0.04
Qwen2.5-1.5B	+1.16	−2.03†	**−1.59**†	Yi-1.5-6B	+0.97	−0.18	**−1.89**†
StableLM-2-1.6B	+1.16	−0.28	−0.15	Qwen2.5-7B	**+3.04**†	+0.09	**−1.17**†
SmolLM2-1.7B	**+2.20**	−0.74	+0.30	Falcon3-7B	−0.26	+0.65	**−1.23**†
Gemma-2-2B	+1.36†	−1.38	**−1.51**†	Mistral-7B-v0.3	**+4.13**†	−1.20	**−2.08**
Granite-3.1-2B	**+1.49**†	+2.12	−0.89	OLMo-2-7B	+1.16	−1.66†	**−2.04**†
Qwen2.5-3B	+0.84	+0.92	−0.85	Gemma-2-9B	+1.29	−1.66	−0.85
Falcon3-3B	+0.39	−0.37	**−2.25**†				

testing the null hypothesis that b and c are drawn from the same underlying 50/50 rate (i.e., among discordant rows, it is a coin flip which rewrite broke it).

Non-parametric bootstrap check. Because that chi-square approximation can be unreliable when $b + c$ is small, every result is cross-checked with a paired bootstrap [20]: the aligned rows are resampled with replacement 10,000 times, $b - c$ is recomputed on each resample, and a 95% percentile interval plus a two-sided empirical p-value are read off the result. A result counts as robustly significant only when both checks agree; when they disagree the more conservative verdict is preferred.

The same sole-cause statistics are additionally pooled across all evaluated models for a given dataset (summing b and c, and resampling from the concatenated pooled rows for the bootstrap), with the explicit caveat that this pooling is not a rigorous mixed-effects test, since different models are not strictly exchangeable independent draws, and is reported as indicative rather than definitive.

3 Experimental results

The sweep covers 51 (model, dataset) runs — all 17 models on each of the three datasets — and so 153 variant-level inference passes, about 34 hours of accelerator time. Every number below is computed over *aligned triples*: rows for which the original and both validated rewrites exist and the model produced a scored prediction on all three, leaving 1,548 rows for SST-2, 1,085 for MMLU and 4,709 for MNLI. Stage 3 also evaluates the full original files (2,000 / 1,107 / 5,127 rows), but those rows have no counterpart to pair against.

The effect is task-dependent, not general (Table 2). On MNLI the hypothesis holds — Δ_{idi} averages −1.16 pp, significantly negative for 10 of 17 models and positive for none, with pooled sole-cause counts of $b = 3{,}351$ against $c = 2{,}751$ ($p < 10^{-4}$ on both tests). On MMLU it holds weakly and only in aggregate, where just one model reaches individual significance but the pooled test does ($p = 0.0008$ McNemar, $p = 0.0012$ bootstrap). On SST-2 it is contradicted: the mean is +1.31 pp, five models are significantly *positive* and none negative, and the pooled sole-cause test is significant in the opposite direction. On MNLI, moreover, idiomaticity costs more than the rewording carrying it (−1.16 against −0.69 pp), whereas on SST-2 the two nearly cancel.

SST-2: rewording hurts, idioms recover. Original accuracy is high and tightly clustered (83.8–91.8%), and plain paraphrasing costs accuracy for 14 of 17 models, significantly for 6. Idiomatizing that same rewording recovers most of the loss — positive for 15 of 17 models, significantly so for 5 — with Mistral-7B-v0.3 showing the largest swing, −2.91 pp from paraphrasing and +4.13 pp from idiomatizing. The parse-failure control below shows, however, that part of this result is artifact rather than comprehension.

Table 4: Worked disagreement cases: three rows on which a model with a *significant* Δ_{idi} on that dataset predicted differently under the paraphrase and the idiomatic rewrite. Each block gives the model's actual prediction for all three variants of one row (✓ correct, × wrong). **Bold** marks the aligned span each variant puts in the same role. Each variant was scored under the same fixed system prompt — MNLI: *"You are a natural language inference classifier … respond with exactly one word: 'entailment', 'neutral', or 'contradiction'."*; SST-2: *"… respond with exactly one word: either 'positive' or 'negative'."*; MMLU: *"… respond with exactly one letter: A, B, C, or D."* — followed by the user prompt shown, so the only thing that differs within a block is the rewritten field. The MNLI and MMLU blocks show the majority direction, where the idiom alone breaks a correct answer; the SST-2 block shows that dataset's reversal, where the plain paraphrase breaks the answer and the idiom restores it.

Variant	User prompt	Prediction
MNLI — Falcon3-3B-Instruct ($\Delta_{\text{idi}} = -2.25$ pp). Every prompt ends `Hypothesis: She now knows what was going on in her mind. / Answer:`. Gold: *entailment*.		
original	`Premise:` It took me really a long time to **sort** what was going on in my mind **out**, she said.	entailment ✓
paraphrase	`Premise:` She stated that it required a significant amount of time for her to **organize her thoughts**.	entailment ✓
idiomatic	`Premise:` It took me quite a while to **wrap my head around** what was going on in my mind, she said.	contradiction ×
SST-2 — Mistral-7B-Instruct-v0.3 ($\Delta_{\text{idi}} = +4.13$ pp). Gold: *positive*.		
original	`Sentence:` is n't embarrassed to **make you reach for the tissues**	positive ✓
paraphrase	`Sentence:` is not ashamed to **make you cry**	negative ×
idiomatic	`Sentence:` isn't shy about **pulling at your heartstrings** and making you reach for the tissues	positive ✓
MMLU — H2O-Danube3-4B-Chat ($\Delta_{\text{idi}} = -2.49$ pp). The four choices are untouched: `A.` a logical mistake. `B.` an ethical mistake. `C.` a factual mistake. `D.` not a mistake. Gold: *D*.		
original	`Question:` **According to Nagel**, the view that moral luck is paradoxical is:	D ✓
paraphrase	`Question:` **Based on Nagel's analysis**, the perspective that moral luck is a paradox is:	D ✓
idiomatic	`Question:` **In Nagel's eyes**, the view that moral luck is paradoxical is:	A ×

MMLU: a small effect that survives only pooling. MMLU spreads models much more widely (29.7–72.3%, against a 25% chance floor) and both deltas are small: Δ_{par} averages -0.07 pp with no model significantly degraded, Δ_{idi} -0.71 pp with only H2O-Danube3-4B individually significant. The sole-cause direction is consistent enough ($b > c$ for 12 of 17 models) that pooling turns it into a clear signal. The small magnitude is likely structural: only the stem is rewritten, while the answer choices — where an MMLU item's knowledge requirement sits — are untouched, so an idiom can obscure what is asked but not what the model must know. Models near the chance floor also generate few sole-cause events (OLMo-2-1B yields $b = 8$, $c = 11$), leaving little power there.

MNLI: consistent idiom-specific degradation. MNLI is the largest evaluation and the one where the manipulation bears directly on the decision. Ten models show a significant penalty, from Falcon3-3B at -2.25 pp to Qwen2.5-7B at -1.17 pp (Table 3), and the sole-cause test agrees: $b > c$ for 15 of 17 models, both tests concurring at $p < 0.05$ for 9 of them — the same models bar Mistral-7B-v0.3, whose accuracy delta is significant while its sole-cause gap ($b = 225$, $c = 191$) is not. Agreement between two tests with different evidence bases across nine independent models is the strongest single piece of evidence we obtain, and it matches the design argument of §2: an entailment judgement turns on a literal reading of the premise, so a figurative surface adds a resolution step that can fail. Table 4 works one such row through in full for a significant model on each dataset.

Robustness to generation-format failures. An unparseable generation is scored as incorrect, so a variant provoking more format violations is penalised for a reason unrelated to idiom comprehension. Rates are low on average (1.3% on SST-2, 0.7% on MMLU, 0.3% on MNLI) but uneven — StableLM-2-1.6B fails on up to 10.5% of MMLU items — and on SST-2 unbalanced across variants: the *paraphrase* arm fails more often than the idiomatic arm for 14 of 17 models, inflating Δ_{idi}. Dropping every triple containing a parse failure (the parse-clean columns of Table 2) leaves MMLU and MNLI untouched, all 10 significant MNLI models included, but halves the SST-2 mean from $+1.31$ to $+0.67$ pp and cuts its significant models from 5 to 3. Qwen2.5-7B is the clearest case: an apparent $+3.04$ pp idiom *advantage* becomes -0.21 pp, because its paraphrase arm failed to parse on 6.4% of rows against the idiomatic arm's 2.1%. The direction does not change — even filtered, no SST-2 model shows a significant idiom penalty.

No capability or scale trend. Idiom sensitivity does not shrink as models grow. Spearman correlations between Δ_{idi} and original accuracy are -0.13 ($p = 0.62$) on SST-2, $+0.18$ ($p = 0.49$) on MMLU and -0.29 ($p = 0.26$) on MNLI; against parameter count, $+0.09$, $+0.07$ and -0.24 — all far from significance. Within the Qwen2.5 ladder on MNLI the effect is non-monotone (-0.40, -1.59, -0.85, -1.17 pp at 0.5B / 1.5B / 3B /

7B), and the two largest models in the panel sit inside the degraded group rather than above it. No correlation reaches significance in either direction; the nearest is a control rather than a treatment effect, SST-2 Δ_{par} against accuracy ($\rho = -0.43$, $p = 0.08$), whose sign is what a ceiling effect predicts. Rewriting is also not purely destructive — pooled across models, 526 SST-2, 284 MMLU and 1,957 MNLI rows were wrong on the original but right on both rewrites, which is precisely why the paraphrase control is needed.

4 Discussion

The sweep supports a narrower claim than the one usually assumed. Idiomatic phrasing is not a general accuracy tax on instruction-tuned decoder LLMs: on binary sentiment it did not hurt at all, and on knowledge QA only one model of 17 reached significance. The cost materialises when the figurative surface is *load-bearing for the decision* — exactly the MNLI configuration — and otherwise tracks how directly the rewritten span bears on the answer: MMLU rewrites the stem while the knowledge sits in untouched choices (−0.71 pp), MNLI rewrites the very span the label is computed from (−1.16 pp). SST-2 inverts the scale (+1.31 pp), its target property being one idioms carry rather than obscure. That coupling, not the presence of idioms, is the most useful predictor we can offer.

The magnitudes are modest even at their strongest. A 1–2 pp mean drop on MNLI is real and statistically robust, but it is the same order as the cost of plain rewording rather than an order above it, so idiomaticity is better read as one distinguishable contributor to paraphrase sensitivity than as a separate failure mode. This tempers, without contradicting, the weaknesses reported directly by Tayyar Madabushi et al. [1] on idiomaticity detection and Chakrabarty et al. [2] on figurative understanding; what we measure is the downstream cost surviving after a model applies whatever partial understanding it has. The practical triage rule: work on idiom robustness pays off where figurative language can land on the span that determines the answer, and is largely wasted where the decision is carried elsewhere. The generic finding deserves equal weight — on SST-2 plain paraphrasing cost more than idiomatizing did. Least comfortable is the absence of a scale trend: across an 18-fold parameter range the effect neither shrinks nor grows reliably, so a larger checkpoint should not be expected to bring idiom robustness with it.

4.1 Threats to validity

Cross-dataset magnitudes carry an augmenter confound. MNLI was augmented with `gemini-3.5-flash-lite`, the other two with `gemini-3.6-flash`, and rewrite depth differs accordingly: median token-set overlap with the original is 0.33 on MNLI against 0.39 on SST-2 and 0.54 on MMLU, though dataset and augmenter vary together and cannot be separated post hoc. Within a dataset both arms share one augmenter, so the paraphrase control still absorbs rewrite depth and the contrast stays internally valid; only MNLI's magnitude against the other two is not cleanly comparable.

Validation is incomplete and not independent. Validation covers label preservation and idiom presence/absence, but nothing tests whether a rewrite preserved the original meaning, so a rewrite that drifted semantically while keeping the label plausible could survive — which matters most exactly where the effect was found, since an MNLI premise whose meaning shifts changes the entailment relation outright. The same augmenter LLM both writes and grades the rewrite, so validation is not independent of generation either.

The statistical treatment has two known weaknesses. The parse-failure filter is blunt — it drops 2.8% of SST-2 rows on average but over 12% for two models there, and 15.9% for StableLM-2-1.6B-Chat on MMLU, and dropping rows non-randomly can itself bias a comparison — so filtered and unfiltered readings bracket the truth. The pooled tests also treat models' items as exchangeable when the same row recurs once per model; a mixed-effects model with model as a random effect was not run, so pooled figures — including MMLU's, whose significance rests entirely on pooling — are indicative, not definitive.

Residual dataset artifacts. MNLI merges the matched and mismatched validation splits, mixing five genres absent from training with five present, and the length filter pruned genres unevenly (50% retention on fiction to 89% on `oup`); genre is recorded per row but not analysed. Forty MNLI paraphrase rows (0.8%) are byte-identical to their premise, weakening the control marginally. All three datasets come from public splits that may appear in pretraining corpora, inflating absolute accuracy — though largely orthogonal here, since every reported quantity is a within-row difference.

Scope limits. Every model was evaluated zero-shot, greedily, under a single prompt template per dataset, so few-shot prompting, sampled decoding and prompt-format variation are untested and could interact with idiomaticity; the fine-tuned encoder track is deferred. Coverage has one hole: Llama-3.2-1B-Instruct is gated on Hugging Face and was never run.

Future work. The most valuable next step would sharpen the MNLI finding rather than broaden it: a direct check that rewrites preserve meaning, ruling out semantic drift as an alternative explanation. A mixed-effects re-analysis would then put the pooled claims on a proper footing, and extending to few-shot prompting would establish whether in-context examples recover the loss, as labelled examples do for the fine-tuned encoders of Tayyar Madabushi et al. [1].

5 Code

Code repository: https://github.com/LihuZur/Idiom_degredation

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